



DRAVIDIAN UNIVERSITY

SRINIVASAVANAM: KUPPAM- 517 426
CHITTOOR (Dist) A.P.

No.DUK/C-II(I)/Off Campus Part-Time Ph.D./Resub./Exams/2026.

Date: 22.09.2026

From

The Co-ordinator

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To

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Sir,

Sub : Dravidian University, Examination Section - Adjudication of Ph.D.
(Off Campus Part-Time) Thesis - Observations - Regarding.

Ref : Adjudicator's report dated 17.11.2020.

-:oOo:-

With reference to the above, I am by direction, I herewith sending a copy of the remarks sent by the examiner, advised to revise and resubmit to be made in the thesis. Hence, you are directed to revise and resubmit two copies of thesis with mentioning present year (2026) on the cover page of the thesis accordingly.

This may kindly be treated as **MOST URGENT**.

Yours faithfully,

CO-ORDINATOR

Enclosed: Copy of remarks sent by the examiner.

Detailed Evaluation Report of Ph.D., Thesis:

Name of the Candidate : Mr.ENUGUNTHY RAMANA CHARY
Reg. No. : 01509227002
UNIVERSITY : DRAVIDIAN UNIVERSITY, KUPPAM – 517 425
DEPARTMENT : SCHOOL OF DISTANCE AND CONTINUING EDUCATION
TITLE : “ANTI-HYPERGLYCEAMIC AND ANTI-HYPERLIPIDEMIC
EFFECTS OF PHYSALIS IN ALLOXAN INDUCED
DIABETIC RATS”

Report:

Introduction:

Diabetes a chronic disorder of metabolism, affects millions of people globally and especially Indians as reported, the global anticipated cases would cross over 400 million in the year 2030. The first study on the prevalence of type 2 diabetes done during 1972 and 1975 by the Indian Council of Medical Research and the prevalence was reported to be 2.8% in the rural area and 5.0% in the urban population of over 40 years. Currently, India is the diabetes capital of the world, estimated to be 70 million diabetes cases in 2025 and its effects on kidney failure, blindness and limb amputation. The recent competitiveness in achieving new drug from the derivatives of plant and its biologically active compounds are the most sought for by the pharmaceutical industries all around the world. The World Health Organization has encouraged the use of herbs as an alternate therapy for treating Diabetics. The traditional use of herbs, in treating numerous clinical conditions, was being practiced, since time immemorial in Indian Ayurvedic System. The plant products are valuable source of nutritious food components as well as being a source of valuable biologically active compounds. There are several plants derived compounds are reported to have effective prevention and treatment of several types of diabetic mellitus viz., proteins, peptides, amino acids, phytochemicals, flavonoids, alkaloids, guanidine, glycolipids, glycosides and others in influencing the level of glucose in human body.

The ethonobotanic environment possess a wide diversity of natural products which are biologically active chemical compounds or substances produced as its by products in treating many diseases. The demand for natural medicinal plants to find application in pharmaceutical, cosmetic, agricultural and food industry is growing day by day. *Physalis minima* L. is an annual herb, reported to have several pharmacological importance such as antitumor, cytotoxic, immune-modulatory, antimicrobial, anti-inflammatory and anti-allergic activity. A very limited research work is being carried out on the effects of the chosen plant on its effects in treating alloxan induced Diabetes Milleus rats. Hence the research scholar has selected an important

research topic to isolate and purify bioactive compounds from *Physalis minima* L. and to study their effect on alloxan induced diabetes mellitus in rat.

Aim and Objectives: The aim and objectives of this study was to evaluate the anti-diabetic activity of *Physalis minima* plant extract. The experiments were designed and carried out to fulfill the objectives of the thesis.

Literature Review: The candidate has written exhaustive references related to the disease, topic of research and the review of literature covers all aspects and provides all the relevant details necessary for this research work. The author has missed to quote several references in the reference section, needs improvement and updation.

Scope of the Thesis: Isolation of bioactive compound Physagulin from the plant extract for the therapeutic approach for DM.

Methodology: The methodology section is written appropriately to suit the research topic. The standard methodologies have been properly followed for isolation and identification of bioactive steroids from selected plant *Physalis minima* and the various analytical tools such as TLC profile and column chromatography are used for purification technique. The elute from TLC was further introduced to NMR studies for compound identification with their structure. The experimental animal adult albino rats of Wister strain were used for the study and were approved by the Ethical committee of Kakatiya University. Induction of diabetes was carried out by the administration of Alloxan intraperitoneally, a proven agent of Diabetes and also confirmed through analyses of fasting glucose concentration. Acute toxicity studies were performed by a well planned experimental design and the results were obtained. The results were statistically analysed using one-way ANOVA.

Results and Discussion: The results were depicted with suitable tables, figures and plates. As per the objectives the chapters were developed. The analysis and interpretation of data are scientifically done. The findings have been carefully discussed and compared with similar type of research works carried out elsewhere with proper statistical analysis.

Summary and Conclusion sections are focused with future needs and for the betterment of the society. The researcher has followed the scientific method of writing. Organization of the thesis is also good. The research scholar has so far not published any research articles.

Findings:

1. The percentage of yield in methanol extract of *Physalis minima* is maximum compared with other four solvents.
2. TLC fraction in the Experimental group IV has yielded single compound in the crude extract and their characterization were carried out by spectral analysis and found to have greater antibacterial activity.
3. The single compound isolated was identified to be Physagulin-F from the fruits of *P. minima* by the Proton and Carbon NMR spectral data analyses.

4. The scholar has concluded that identified compound: Physagulin-F found to have Anti-hyperglycaemic and anti-hyperlipidemic activities.
5. Further research on biochemical and pharmacological studies will elucidate the mechanism of action and the plant products will be a therapeutic target in diabetes treatment.

Questions to be asked and clarified:

1. Have you obtained the authentication report of the chosen plant *Physalis minima* L. from the Botanical Survey of India or else regional plant taxonomist? If so provide a copy of the same.
2. Why the author have not enclosed any photograph of the TLC plates detailing the separation.
3. The author claims to have done NMR studies for compound identification, but the chromatogram was not enclosed. Why?
4. Has the thesis been checked for plagiarism by the University
5. ANOVA study was carried out by means of which software.
6. There is no evidence of antimicrobial activity plates.
7. References many cited in the text were missing in the Reference section. Include.
8. Has the author fulfilled the requirement of publication of his work in scientific journals for submission of thesis?

Recommendation: Based on the originality of the research work carried out by ENUGUNTHY RAMANA CHARY (Reg.No.:01509227002), School Of Distance And Continuing Education "Anti-Hyperglycaemic and Anti-Hyperlipidemic effects of Physalis in Alloxan induced diabetic rats" is advised to revise and resubmit the thesis carrying out the above mentioned clarifications.